

Individual Contribution Assessment

Student Name: Nidal El Ibrik - **Student ID:** 6945 - **Project Title:** Smart Solar Tracking Self-Cleaning System

My primary responsibility in this project was the preparation and development of the final technical report. I independently wrote and organized the complete report, including its technical explanations, experimental methodology, results, discussions, references, figures, tables, graphs, photographs, captions, and data sheets. I was also responsible for ensuring that the report accurately represented the final physical prototype and its actual capabilities. This required continuously comparing the written system description with the implemented hardware and Arduino code and correcting any inconsistencies that were identified.

In addition to report preparation, I contributed significantly to the original project concept and system planning. I participated in defining the main purpose of the system and proposed several of its components and functions. These included dual-axis solar tracking, automated water pumping, water-level monitoring using a load cell, environmental measurements, LCD information display, and safety conditions for the cleaning system. I also participated in component selection and helped ensure that the selected sensors and actuators could operate together as an integrated system.

During implementation, I worked with the group on the mechanical construction, assembly, hardware integration, programming, and testing of the prototype. My direct electrical contributions included assisting with the load-cell and pump connections. I also helped modify and combine existing Arduino code so that the different system functions could operate together. Because I worked closely with the calibration procedures and experimental data, I contributed to selecting suitable thresholds, delays, and operating conditions, particularly for pump control and safety-related functions.

I participated in calibrating the load cell by measuring known masses using an external scale, placing the same masses on the load cell, and recording the corresponding sensor readings. The collected data were then used to produce a linear calibration model that converted the load-cell output into estimated mass. I also participated in solar-tracking tests, pump-control tests, LCD display verification, and the testing of safety features, including preventing pump operation when the available water quantity was insufficient.

Several practical problems were encountered during the project, including inconsistent LDR readings, unstable load-cell and temperature measurements, unreliable flow-sensor readings, and pump-control problems in which the pump either remained continuously active or failed to operate. I assisted the team in troubleshooting these problems by checking wiring, reviewing the code, modifying thresholds and delays, repeating measurements, and comparing the observed results with the expected system behavior.

I additionally reviewed the final report and code for alignment. Some operating thresholds differed between the report and the Arduino program and were corrected. Certain initially proposed functions also required revision after experimental testing. For example, the infrared dust sensor was initially investigated as a continuous analog measurement device. However, its output was strongly influenced by environmental conditions and required frequent recalibration. Therefore, the final system description was revised to represent it more accurately as a threshold-based detection method rather than a consistently quantitative dust-measurement system.

Finally, I contributed to documenting the completed prototype by recording the demonstration video and explaining the main circuit components at the beginning of the recording.

1. Personal Reflection

This project demonstrated that theoretical system concepts can appear simple when studied individually but become significantly more difficult when implemented using physical hardware. A solar tracker, pump-control system, or load-cell measurement procedure may operate clearly in theory; however, real sensors are affected by noise, wiring quality, environmental conditions, mechanical alignment, power stability, and calibration errors. The project therefore required considerably more patience, repetition, and troubleshooting than initially expected.

One of the most important technical lessons I learned was that sensor calibration is not always permanent. Readings that appeared acceptable under one set of environmental conditions could change after the sensor position, lighting conditions, wiring, load distribution, or surrounding environment was modified. This was particularly noticeable with the load cell, LDRs, temperature measurements, flow sensor, and infrared dust sensor. I learned that sensors are highly sensitive devices and that reliable measurements require proper mounting, stable wiring, repeated testing, and clearly defined calibration procedures.

The most challenging parts of the project for me were calibration and data analysis. These tasks required converting unstable or indirect sensor readings into meaningful physical quantities and then determining whether the resulting relationship was sufficiently reliable for use in the system. To address these difficulties, I consulted technical resources, used available online troubleshooting tools, repeated the experiments, compared measurements, and discussed possible solutions with the other group members.

The unsuccessful attempt to obtain a consistently reliable analog dust measurement from the infrared sensor provided one of the most valuable lessons. Although data were collected and analyzed, the sensor response changed considerably with environmental conditions. Reliable continuous measurement would therefore have required recalibration and new threshold selection whenever the conditions changed. This showed me that an unsuccessful experiment can still provide an important engineering conclusion. Instead of presenting the sensor as more accurate than it was, the report and system description were revised to represent its practical limitations honestly.

The project also emphasized the importance of communication and collaboration. Since several tasks were completed simultaneously, progress depended on the group members communicating hardware changes, code modifications, testing results, and system limitations. Effective communication made it easier to combine the separate parts into one functioning prototype and helped prevent contradictions between the physical system, code, and report.

If the project were repeated, I would begin with a more detailed implementation and testing plan. The group initially experienced some confusion regarding the system structure and the order in which components should be integrated. I would divide the system into smaller subsystems, establish the final pin assignments and operating logic earlier, test every subsystem independently, and document each change immediately. Overall, the skills that improved most were technical report writing and sensor calibration. The project strengthened my ability to interpret experimental results, identify limitations, and present engineering work accurately rather than describing only the intended theoretical performance.